Tabletop Traffic Junction

Observe, decide, actuate—without conflicting greens

The signal heads reveal the state machine; the display reveals its time.

Lesson	012 — integration project	**Time**	180–240 minutes
Prerequisites	Lessons 001–011	**Board**	Arduino Mega 2560
Evidence	two signal heads, pedestrian pair, countdown, test points	**Status**	host verified; physical acceptance open
Safety	E1 tabletop model; USB-powered current-limited LEDs only		

Safety checkpoint. This model is not a road controller and must never direct real traffic. Disconnect USB before wiring. Every LED and display segment needs its own resistor. Use only the Mega's 5 V rail. Stop for heat, odor, resets, unexpected simultaneous greens, or disagreement between the schematic and breadboard. Remove USB power to stop the experiment.

1 Mission and evidence chain

Build two vehicle signal heads, a pedestrian request and signal pair, and a one-digit countdown. The project composes:

button → `TrafficInput` → `TrafficJunction` → `TrafficSignals` → current-limited LEDs

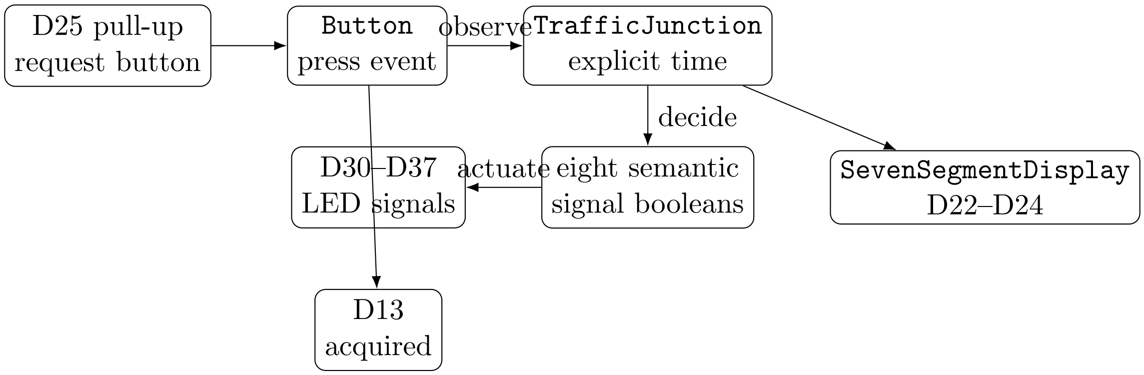
`nextDeadline` → `SevenSegmentDisplay` → `ShiftRegisterOutput` → visible digit.

The D13 board LED turns on only after every first-class object initializes, so it is the acquisition indicator. Initialization is transactional: a failed step shuts down every earlier object in reverse dependency order. The heads are the primary diagnostic channel. All-red plus pedestrian stop is the requested running safe state. The countdown is independent supporting evidence: it shows the remaining whole seconds until the next transition. Serial is not required. Named electrical test points are D30 (main red), D32 (main green), D33 (side red), and D35 (side green), each measured relative to GND.

By the end, you can:

- explain why a hardware-neutral state engine receives explicit time;
- trace a latched pedestrian request until it is served;
- prove from observations that both greens are never active together;
- explain why actuation turns permissive signals off before changing red;
- relate a semantic glyph to an atomic 74HC595 latch update;
- distinguish successful acquisition, a commanded all-red state, and a measured electrical all-red state.

2 System composition



The canonical sketch follows the same narrative. `loop()` observes one button event, asks the engine to decide, then actuates the returned snapshot. It never invents signal combinations in Arduino-facing code.

2.1 The state promise

Phase	Vehicle indication	Pedestrian indication
`AllRed`	both red	stop
`MainGreen/MainYellow`	named main aspect; side red	stop
`SideGreen/SideYellow`	named side aspect; main red	stop
`PedestrianWalk`	both red	walk
`PedestrianClearance`	both red	stop
`Fault`	both red	stop

A request is an event, not a held electrical level. The engine latches an accepted request and serves it after the side phase. Each observation also checks that every owned endpoint remains initialized. That check detects lost software ownership; it is not electrical feedback from an LED. A reported circuit-health failure latches `Fault`. The sketch then requests all-red best-effort and performs canonical shutdown. No call blocks while a phase elapses.

3 Orientation drawing

4 Parts, limits, and literal wiring

Quantity	Part	Requirement
1	Mega 2560	USB powered
1	74HC595	identified pinout; 0.1 μF local decoupling
1	one-digit display	common cathode, identified segment pinout
8	signal LEDs	two red/yellow/green heads plus red/green pedestrian pair
16	resistors	eight 330 Ω signal; eight 1 kΩ display
1	momentary button	normally open
1	multimeter	unpowered continuity and powered DC voltage

Mega and signal connections.

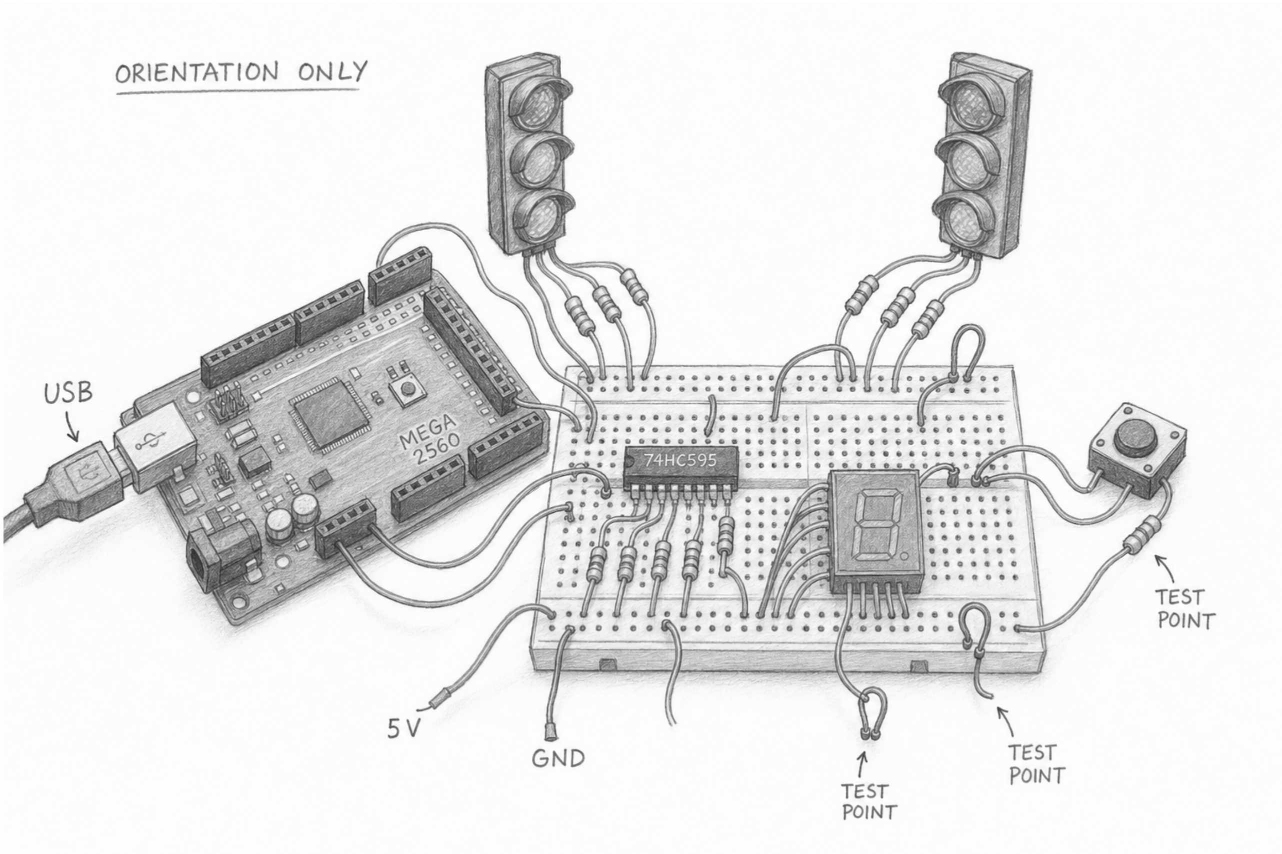


Figure 1: Pencil orientation of the Mega, two model heads, request button, shift register, display, and test points. It is not an authoritative wiring diagram. Breadboard connectivity and package pin order are simplified; use the literal connection table and each part's identified datasheet.

Function	Mega pin	Literal path
request	D25	D25 to button to GND; internal pull-up
acquisition	D13	built-in LED; on only after every object initializes
main head	D30/D31/D32	pin to 330 Ω to red/yellow/green anode; every cathode to GND
side head	D33/D34/D35	same order and resistor rule
pedestrian	D36/D37	walk green/stop red, each through 330 Ω

74HC595 and display connections.

Function	Connection	Required interpretation
serial data/clock/latch	D22 to DS, D23 to SHCP, D24 to STCP	`ShiftRegisterPins{22,23,24}`
control	MR to 5 V; OE to GND	never leave either floating; see startup limit below
power	VCC to 5 V; GND to GND; 0.1 μF across them	place capacitor at the IC
segments	Q0–Q6 to a–g, Q7 to decimal point	each path has its own 1 kΩ resistor
display return	both common cathodes to GND	verify the exact package pinout

Do not copy a display pin number from the pencil drawing. Identify the part and check continuity unpowered. With a conservative 3 mA per lit display segment and 10 mA per signal LED, the worst expected visible state is under 55 mA; record actual resistor values, forward voltages, and measured current before claiming that budget.

Safety checkpoint. With OE tied low, the 74HC595 storage register can present an unspecified pattern before the first software latch. The three-wire lesson circuit therefore cannot promise a blank display at power-up or reset. Its atomic-update claim begins only after initialization. A later hardened adapter must own OE and hold it disabled with a pull-up until a known blank value has been latched.

5 Build and run from the CLI

```
make arduino-Lesson012TrafficJunction
make upload-Lesson012TrafficJunction PORT=/dev/ttyACM0
make monitor-describe PORT=/dev/ttyACM0
make serial-log PORT=/dev/ttyACM0 \
    SERIAL_LOG=build/evidence/lesson012-optional.log
```

Serial capture is optional context. The required proof is the visible and electrical evidence below. Build and upload targets must remain the canonical path; no IDE-specific step is part of the lesson.

5.1 The narrative run loop

Listing 1: Canonical observe–decide–actuate flow.

```
    void                 haltJunction     ();
} // namespace

void setup ()
{
    halted = !acquireJunction ();
}

void loop ()
{
    if (halted)
    {
        return;
    }

    const adk::TimePoint       now                           (millis ());
    const adk::TrafficInput    observation = observeJunction   (now);
    const adk::Status          status      = decideJunction    (now, observation);
    const adk::TrafficSnapshot decision    = junction.snapshot ();

    if (!status.ok () || !actuateJunction (now, decision))
    {
        haltJunction ();
```

The engine's `TrafficSignals` is the sole actuator model. The `showSignals()` adapter first removes green, walk, and yellow permission; then it establishes all-red; only then does it present the requested state. This ordering reduces transition hazards when writes succeed. It cannot guarantee an electrical state after a driver, wire, LED, or supply failure. The fault path attempts every all-red write even if an earlier write fails, blanks the display, and releases every endpoint to high impedance.

6 Predict, observe, interpret

Predict	Observe	Interpret
Acquisition completes	D13 turns on	every project object initialized; this does not prove the signal pin levels
Startup is all-red	both red heads and pedestrian stop illuminate before any green	the first semantic state was actuated; measure test points before calling it electrical proof
Only one vehicle direction proceeds	watch an entire main/side cycle; compare D32 and D35	any simultaneous high reading is a stop condition and a failed acceptance test
One request is remembered	press and release D25 once during main green; wait without holding it	a later pedestrian walk proves event capture and latching
The display changes atomically	watch the countdown at a phase boundary	mixed or torn segments indicate wiring, resistor, or latch trouble
Engine fault requests all-red	use the host fault fixture; do not create shorts on the powered circuit	the snapshot commands two reds and pedestrian stop; adapter failure only makes a best-effort request before shutdown
Shutdown is electrically inactive	upload a dedicated acceptance sketch that calls endpoint shutdown; measure each named pin	high impedance is distinct from the running model's commanded all-red state

Do not merge three claims. D13 proves completed acquisition. D30/D32/D33/D35 measurements establish the commanded electrical levels. A separate shutdown measurement establishes endpoint high impedance. Record all three independently. Software cannot prove that a lamp or wire actually followed its command; only observation can.

7 Deterministic desk experiment

Before wiring, replay this trace on paper. Use the configured phase durations from the canonical source.

1. At $t = 0$, initialize. Record phase, all eight signals, and deadline.
2. Update immediately before and exactly at the startup deadline.
3. During main green, supply one `pedestrianRequestEvent=true`. Return it to false on the next update.
4. Advance exactly to each deadline through main yellow, clearance, side green, side yellow, clearance, walk, and pedestrian clearance.
5. At every row, evaluate $\neg(\texttt{mainGreen} \wedge \texttt{sideGreen})$.
6. Repeat the identical trace. Snapshots must match byte for byte.

Time	Input event	Expected phase	Main	Side	Pedestrian/pending
________	________	________	________	________	________
________	________	________	________	________	________
________	________	________	________	________	________
________	________	________	________	________	________

8 Diagnosis

Observation	Likely layer	Next bounded check
No LEDs and blank display	power or acquisition	remove USB; inspect rails, polarity, pin map, and resource conflicts
Heads correct, digit wrong	display encoding/wiring	run lesson 010; compare Q0–Q7 with a–g/decimal point
Digit tears during change	latch wiring	inspect STCP D24 and shared ground; do not add delays to hide it
Request never served	button/event path	observe raw D25 and repeat lesson 003 debounce evidence
Both greens visible	actuator wiring or severe defect	remove USB immediately; unpowered continuity from D32/D35 to the intended LEDs
All-red forever	startup/fault/configuration	reproduce the same timestamps in host tests and inspect `status`
Unexpected resets	current or wiring	remove USB; measure resistors and inspect for rail shorts before reapplying power

9 Exercises

1. Add a host trace with a request one millisecond before a deadline. State exactly which snapshot accepts it and why.
2. Prove that every path from one green direction to the other contains `AllRed` for at least `vehicleAllRed`.
3. Replace the numeric countdown with `Dash` during fault. Keep the heads as the authoritative safe-state evidence.
4. Design a maintenance self-test that illuminates one aspect at a time. It may run only from all-red and must return to all-red on every error.
5. Calculate measured LED current from resistor voltage, then compare the sum with your predicted current budget.

10 Bench acceptance record

This source has host and compile evidence only. Physical acceptance is deferred until a completed, reviewed record is committed.

Date	____________	Reviewer	____________
Board/revision	____________	Supply/USB source	____________
Meter	____________	Firmware commit	____________
74HC595 marking	____________	Display marking	____________

- ☐ Unpowered continuity matches every literal connection.
- ☐ Every signal resistor measures ________; every segment resistor measures ________.
- ☐ Startup visibly reaches all-red before a green.
- ☐ D13 turns on only after full acquisition.
- ☐ D32 and D35 are never simultaneously high over three cycles.
- ☐ One short D25 press is eventually served exactly once.
- ☐ The digit changes without a visible intermediate pattern.

- ☐ Maximum measured signal current is ________ mA; display current is ________ mA; total is ________ mA.
- ☐ A healthy output path presents the requested running fault indication: all-red plus pedestrian stop.
- ☐ Injected adapter failure attempts every safe-state write, then performs canonical reverse-order shutdown.
- ☐ Separate shutdown firmware leaves output pins measured high impedance.
  Method/result: ______________________________.
- ☐ Reset and USB removal produce no observed conflicting green.

Deviations and observations:

11 What to carry forward

The reusable idea is not "a traffic light." It is the boundary between a deterministic decision and a transactional physical presentation. The engine speaks in domain signals; endpoint-owning adapters perform safe ordering; the circuit supplies evidence alongside its useful behavior. The same separation supports later environmental, access, rover, greenhouse, telemetry, and inert cue projects.

Primary source paths: Traffic junction API, seven-segment API, shift-register API, and canonical example.